

Miscellaneous — Tricks, Traps, Counterexamples, Synthesis

The exam-night revision sheet

This summary catalogues what's easy to forget, easy to confuse, and easy to misapply across the entire single-variable calculus / real analysis arc. It is organised by failure mode and synthetic theme rather than by topic, so cross-topic connections become visible.

1 The counterexample library

These are the functions worth memorising — each disproves something a student might otherwise believe.

Function	What it disproves / what to remember
$f(x) = x $	Continuous everywhere, not differentiable at 0.
Weierstrass function $\sum a^n \cos(b^n \pi x)$	Continuous everywhere, differentiable nowhere.
$f(x) = x^3$	Strictly increasing despite $f'(0) = 0$. So f strictly $\uparrow$ does <i>not</i> imply $f' > 0$ everywhere.
$f(x) = x^4$	$f''(0) = 0$ but 0 is a local min, not an inflection point. So $f''(c) = 0$ alone doesn't give an inflection.
Dirichlet function: $\mathbb{K}_{\mathbb{Q}}$	Discontinuous everywhere, not Riemann integrable, has no antiderivative.
Thomae (popcorn) $f(p/q) = 1/q, f(\text{irr}) = 0$	Continuous exactly at irrationals, Riemann integrable, integral is 0.
$\sin(1/x)$ on $\mathbb{R} \setminus \{0\}$	Essential discontinuity at 0; $\lim_{x \rightarrow 0}$ does not exist.
$x \sin(1/x)$ extended by 0	Continuous at 0; not differentiable at 0.
$x^2 \sin(1/x)$ extended by 0	Differentiable at 0 with $f'(0) = 0$; but f' is essentially discontinuous at 0.
Heaviside step $H(x)$	Cannot be the derivative of any function (Darboux: derivatives can't have jumps).
$f(x) = e^{-1/x^2}$ extended by 0	C^∞ at 0, all derivatives at 0 are zero, yet $f \not\equiv 0$: Taylor series at 0 gives 0, missing the function.
$f_n(x) = n\mathbb{K}_{(0,1/n]}$ on $[0, 1]$	Pointwise $\rightarrow 0$ but $\int f_n = 1$: shows $\lim \int \neq \int \lim$ without DCT/MCT hypotheses.
$f_n(x) = x^n$ on $[0, 1]$	Pointwise $\rightarrow 0$ on $[0, 1)$ and $\rightarrow 1$ at 1; pointwise limit not continuous, even though each f_n is.
$1/n^p$ series with $p = 1$ vs $p = 2$	$\sum 1/n$ diverges, $\sum 1/n^2$ converges. The boundary is exactly $p = 1$.
$\sum (-1)^{n+1}/n = \ln 2$ vs $\sum 1/n = \infty$	Conditional convergence is delicate: rearrangement can give any real number (Riemann).

2 Theorem-hypothesis traps

For every major theorem, it pays to know exactly which hypothesis fails in which counterexample. The exam frequently asks “which of the following *must* hold,” rewarding a student who can spot a dropped hypothesis.

2.1 Continuity-related theorems

IVT — needs continuity on a connected interval.

- Drop continuity: $f(x) = -1$ for $x < 0$, $f(0) = 0$, $f(x) = 1$ for $x > 0$ has $f(-1) = -1, f(1) = 1$ but never takes value $1/2$.
- The interval must be connected (an interval, not a union of intervals).

EVT — needs continuity, closed interval, bounded interval.

- Drop continuity: $f(x) = 1/x$ on $(0, 1]$ extended by $f(0) = 0$ is unbounded, no max.
- Drop closed: $f(x) = x$ on $(0, 1)$ has $\sup = 1$ and $\inf = 0$, neither attained.
- Drop bounded: $f(x) = x$ on $[0, \infty)$ has min at 0 but no max.

Heine–Cantor (uniform continuity) — needs closed and bounded. $f(x) = 1/x$ on $(0, 1]$ is continuous but not uniformly continuous. $f(x) = x^2$ on $[0, \infty)$ is continuous but not uniformly continuous.

2.2 Differentiation-related theorems

Rolle — needs continuity on $[a, b]$, differentiability on (a, b) , $f(a) = f(b)$.

- Drop differentiability: $f(x) = |x|$ on $[-1, 1]$ has $f(-1) = f(1)$ but f' never zero (it's ± 1 off 0, undefined at 0).
- Drop endpoint equality: $f(x) = x$ on $[0, 1]$, no critical point.

MVT — same hypotheses as Rolle (without endpoint equality). The same $f(x) = |x|$ on $[-1, 1]$ shows MVT can fail without differentiability: chord slope is 0, but no point in $(-1, 1)$ has $f'(c) = 0$.

L'Hôpital — needs $\frac{0}{0}$ or $\frac{\infty}{\infty}$, $g' \neq 0$ near a , AND $\lim f'/g'$ to exist.

- Failure when $\lim f'/g'$ doesn't exist: $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x} = 1$ (by squeeze), but $\frac{(x + \sin x)'}{x'} = 1 + \cos x$ has no limit. L'Hôpital is silent here, not wrong.
- Forgetting to check the form is indeterminate: $\lim_{x \rightarrow 0} \frac{x}{1+x} = 0$, not what L'Hôpital would give.

FTC Part I — needs f continuous on $[a, b]$ for $\int_a^b f = F(b) - F(a)$ via any antiderivative. Without continuity, an antiderivative might not even exist.

FTC Part II — needs f continuous at x for $G'(x) = f(x)$. If f has a jump at x_0 , $G(x) = \int_a^x f$ is still continuous at x_0 but $G'(x_0)$ might not equal $f(x_0)$.

2.3 Convergence-theorem traps

DCT — needs a single integrable dominator g . Without a dominator, $\lim \int \neq \int \lim$ as the $n\mathbb{K}_{(0,1/n]}$ example shows.

MCT — needs the sequence to be monotone increasing AND non-negative. A decreasing sequence $f_n = -1 \cdot \mathbb{K}_{(0,1/n]}$ doesn't fit; nor does an oscillating one.

Term-by-term differentiation/integration of series — needs uniform convergence (of the differentiated series, not just the original). A series $\sum f_n$ converging pointwise doesn't allow swapping $\sum$ with $\int$ or d/dx in general.

3 Growth-rate hierarchy and standard limits

3.1 The asymptotic ladder

For any $\alpha, \beta, \gamma > 0$:

$$(\ln \ln x)^\alpha \ll (\ln x)^\alpha \ll x^\beta \ll e^{\gamma x} \ll x^x \ll e^{x^2} \ll \dots$$

where $\ll$ means “grows much slower than as $x \rightarrow \infty$.”

Quantitative statements.

- $\lim_{x \rightarrow \infty} \frac{(\ln x)^\alpha}{x^\beta} = 0$ for any $\alpha, \beta > 0$.
- $\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$ for any n (polynomial vs exponential).
- $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$ for any $a > 0$ (exponential vs factorial).
- $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$ (factorial vs n^n , with rate $\sim \sqrt{2\pi n} \cdot e^{-n}$).
- $\lim_{n \rightarrow \infty} n^{1/n} = 1$ (root growth $\rightarrow 1$).
- $\lim_{n \rightarrow \infty} (n!)^{1/n}/n = 1/e$ (Stirling).

3.2 The 12 standard limits worth memorising

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$\lim_{x \rightarrow 0^+} x \ln x = 0$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\arcsin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\lim_{x \rightarrow 0^+} x^x = 1$$

4 Common MCQ baits

Tempting wrong belief	The truth
f continuous + injective on interval $\Rightarrow f$ is monotone	<i>True!</i> (Often misremembered as “not necessarily.”) Easy to forget.
f continuous on $\mathbb{R}$ + bounded $\Rightarrow$ attains max	False: $f(x) = \arctan x$ doesn't attain $\pm\pi/2$. EVT needs closed bounded interval.
f differentiable $\Rightarrow f'$ continuous	False: $x^2 \sin(1/x)$ extended by 0.
f' exists everywhere $\Rightarrow f'$ has IVP	<i>True</i> (Darboux). But f' can be highly discontinuous.
$f' \geq 0$ on $I \Rightarrow f$ strictly increasing on I	False if equality on a sub-interval. True if $f' > 0$ on I (or ≥ 0 with f' vanishing only on a set with no interior).
f strictly increasing $\Rightarrow f' > 0$ on I	False: $f(x) = x^3$.
$f''(c) = 0 \Rightarrow$ inflection at c	False: x^4 . Need f'' to change sign at c .
$f''(c) > 0$ AND $f'(c) = 0 \Rightarrow$ local min	True.
$f''(c) = 0$ AND $f'(c) = 0 \Rightarrow$ no extremum	False: x^4 has min at 0. Use higher derivatives or 1st-derivative test.
$\lim f_n = f$ pointwise + f_n continuous $\Rightarrow f$ continuous	False: x^n on $[0, 1]$. Need uniform convergence.
$\sum a_n$ converges $\Rightarrow a_n \rightarrow 0$	True.
$a_n \rightarrow 0 \Rightarrow \sum a_n$ converges	False: $\sum 1/n$.
$\sum a_n $ converges $\Rightarrow \sum a_n$ converges	True.
$\sum a_n$ converges $\Rightarrow \sum a_n $ converges	False: $\sum (-1)^n/n$.
f Riemann integrable $\Rightarrow f$ continuous a.e.	True (Lebesgue).
f Riemann integrable $\Rightarrow f$ continuous somewhere	True (consequence). But Thomae shows discontinuity at every rational is allowed.
Continuity exactly on $\mathbb{Q}$ is achievable	False: $\mathbb{Q}$ is not a G_δ set.
Continuity exactly on irrationals is achievable	True: Thomae's function.
Derivative can have a jump discontinuity	False: Darboux forbids it.
Derivative is bounded $\Rightarrow f$ is Lipschitz	True (MVT).
f Lipschitz $\Rightarrow f$ differentiable everywhere	False: $ x $. (But Lipschitz $\Rightarrow$ differentiable a.e. by Rademacher.)

5 Inequality cheatsheet (cross-topic)

5.1 Tangent-line / convexity inequalities

- $e^x \geq 1 + x$ for all x , equality iff $x = 0$.
- $\ln(1+x) \leq x$ for $x > -1$, equality at 0.
- $\frac{x}{1+x} \leq \ln(1+x) \leq x$ for $x > -1$.
- $\sin x \leq x$ for $x \geq 0$; $\sin x \geq x - \frac{x^3}{6}$ for $x \geq 0$.
- $\cos x \geq 1 - \frac{x^2}{2}$ for all x .
- $1 - x \leq e^{-x}$ for all x .
- $|\sin x - \sin y| \leq |x - y|$, $|\cos x - \cos y| \leq |x - y|$ (Lipschitz with constant 1).

5.2 Mean inequalities

For positive $a_1, \dots, a_n$:

$$\underbrace{\frac{n}{\sum 1/a_i}}_{\text{HM}} \leq \underbrace{\sqrt[n]{\prod a_i}}_{\text{GM}} \leq \underbrace{\frac{\sum a_i}{n}}_{\text{AM}} \leq \underbrace{\sqrt{\frac{\sum a_i^2}{n}}}_{\text{QM/RMS}}.$$

Equality throughout iff all a_i are equal.

5.3 Integral inequalities

- **Cauchy–Schwarz:** $(\int fg)^2 \leq (\int f^2)(\int g^2)$.
- **Chebyshev sum:** same monotonicity $\Rightarrow (b-a) \int fg \geq (\int f)(\int g)$.
- **Hermite–Hadamard:** for convex f , $f(\frac{a+b}{2}) \leq \frac{1}{b-a} \int_a^b f \leq \frac{f(a)+f(b)}{2}$.
- **Young (exponents):** $ab \leq a^p/p + b^q/q$ when $1/p + 1/q = 1$, $p, q > 1$.

- **Jensen (integral):** φ convex, $w \geq 0$, $\int w = 1$: $\varphi(\int xw) \leq \int \varphi(x)w$.

5.4 Bounding tricks

- $|\int_a^b f| \leq \int_a^b |f| \leq (b-a) \max_{[a,b]} |f|$.
- $\sin x, \cos x \in [-1, 1]$ always: bound trig factors by 1 and don't lose more than necessary.
- $|\arctan x| < \pi/2$ always.

6 Notational and definitional traps

lim vs lim inf vs lim sup. The limit may not exist while lim inf and lim sup always do (in $[-\infty, \infty]$). Equality $\liminf = \limsup$ is the criterion for lim to exist.

Pointwise vs uniform convergence.

- Pointwise: $\forall x, \forall \varepsilon, \exists N(x, \varepsilon)$. N depends on x .
- Uniform: $\forall \varepsilon, \exists N(\varepsilon), \forall x$. Same N works for all x .
- **Uniform $\Rightarrow$ pointwise**, never the converse.

Continuous on a set vs continuous at every point of the set. For unusual sets (like the Cantor set), these can differ. On standard intervals, they're the same.

"Differentiable" vs "has a derivative". Same thing in a single-variable context. But in MVT, "differentiable on (a, b) " means *everywhere* on (a, b) , not just at one point.

Open vs closed interval matters.

- f continuous on $[a, b]$: bounded, attains max/min, uniformly continuous.
- f continuous on (a, b) : none of these guaranteed.

$\int_a^b$ vs $\int_a^x$. The first is a number; the second is a function of x . FTC Part II says the second is differentiable.

$\Rightarrow$ and $\Leftarrow$. Many theorems are stated as "if-then" with the converse *also* true, but easy to forget. Examples:

- Monotone bounded sequence $\Leftrightarrow$ convergent (one direction is monotone-convergence, other is trivial).
- Cauchy sequence in $\mathbb{R} \Leftrightarrow$ convergent (completeness).
- f Riemann integrable $\Leftrightarrow$ bounded with discontinuities of measure zero.

"Exists" vs "is finite". $\lim_{x \rightarrow 0} 1/x^2 = +\infty$: the limit *exists* in the extended sense but is not a real number. Phrasing matters in MCQs.

7 Cross-topic synthesis

The same idea often appears under different names across topics. Recognising the unity is half the battle.

7.1 The "derivative cannot have a jump" theme

Three ways the same fact appears:

- **Darboux's theorem:** f' has the intermediate-value property.
- **Derivative-limit theorem:** If $\lim_{x \rightarrow a} f'(x) = L$ exists, then $f'(a) = L$.
- **No-jump consequence:** f' 's only allowed discontinuities are essential (oscillatory). Removable and jump types are forbidden.

7.2 The "existence under continuity" theme

- **IVT:** continuous f takes every intermediate value.
- **EVT:** continuous f on closed bounded interval attains max/min.
- **Boundedness:** continuous f on closed bounded interval is bounded.
- **Heine–Cantor:** continuous f on closed bounded interval is uniformly continuous.
- **FTC II:** continuous f has antiderivative $\int_a^x f$.
- Common thread: closed bounded interval + continuity is the standard hypothesis package, repeated everywhere.

7.3 The "Rolle in disguise" theme

Every mean value theorem is Rolle plus a clever auxiliary function:

- Lagrange MVT: apply Rolle to $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$.
- Cauchy MVT: apply Rolle to $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$.
- " $f'(c) = \lambda f(c)$ " problems: apply Rolle to $g(x) = e^{-\lambda x} f(x)$.

7.4 The "convert sum to integral" theme

- Riemann sum: $\frac{1}{n} \sum_{k=1}^n f(k/n) \rightarrow \int_0^1 f$.
- Stolz–Cesàro: discrete L'Hôpital for sequences of ratios.

- Stirling: replace $n!$ by its continuous approximant $(n/e)^n \sqrt{2\pi n}$.
- Integral test for series: $\sum f(n)$ converges iff $\int f$ converges.

7.5 The “differentiate under to evaluate” theme

- Differentiation under the integral sign (Leibniz/Feynman): introduce a parameter.
- L'Hôpital + FTC: differentiate to strip an integration.
- Taylor expansion: differentiate enough times to get a polynomial approximation, then integrate the remainder.

7.6 The “ e^x multiplication” theme

- Rolle on $e^{-x}f(x)$ to get $f' = f$.
- Hardy's trick: $f(x)/1 = e^x f(x)/e^x$, then L'Hôpital.
- Integrating factor for ODEs $f' + pf = q$: multiply by $e^{\int p}$.
- Showcase: $f + f' \rightarrow 0 \Rightarrow f \rightarrow 0$ via e^x multiplication and extended L'Hôpital.

8 Exam survival checklist

Before starting any analysis problem, ask:

1. Is the function continuous? On what set? At what points might it fail?
2. Is the interval closed and bounded? Open? Unbounded?
3. Does the answer depend on the order of limits, integrals, sums, or derivatives?
4. Is there a hidden indeterminate form?
5. Could a counterexample disprove the claim? (Run through the library.)

When proving a positive statement:

1. Verify all hypotheses of the theorem you cite.
2. For inequalities, define $h = \text{LHS} - \text{RHS}$ and chase signs of $h, h', h'', \dots$
3. For existence, look for sign changes (IVT), interior extrema (Rolle/Fermat), or two-sequence arguments.
4. For uniqueness, combine existence with monotonicity (IVT + Rolle).

When evaluating a limit or integral:

1. Identify the indeterminate form.
2. Try the standard substitutions/expansions before reaching for L'Hôpital.
3. For sequence-limits with sums, recognise Riemann sums.
4. For $n!$ in n th-root limits, reach for Stirling.
5. For trig + exponential mixes, replace trig by e^{ix} and take real/imaginary parts.
6. For integrals with parameters, consider differentiation under the integral.

Before circling an answer:

1. Check the dimensions / units (if applicable).
2. Check a special value (e.g., $x = 0, a = b, n = 1$).
3. Check the asymptotic behaviour ($x \rightarrow 0, x \rightarrow \infty$).
4. Check signs — does the answer make sense for the kind of object it represents?

Three universal moves across analysis: (1) *Build an auxiliary function* ($h = f - g, g = e^{-\lambda x}f$, etc.) to convert the problem into something Rolle, MVT, or IVT can attack. (2) *Recognise the canonical form* — Riemann sum, Taylor expansion, integrating factor, e^{ix} replacement. (3) *Exploit hypothesis-failure when stuck* — if a standard theorem doesn't directly apply, identify which hypothesis fails and engineer a workaround (e.g., extend to closure, restrict to a sub-interval, take a limit of valid cases).